

Variability and Energy Consumption Tradeoffs in Multilevel Programming of RRAM Arrays

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Abstract—Achieving a reliable multilevel programming operation in resistive random access memory (RRAM) arrays is still a challenging task. In this work, we assessed the impact of the voltage step value used by the programming algorithm on the device-to-device (DTD) variability of the current distributions of four conductive levels and on the energy consumption featured by programming 4-kbit HfO₂-based RRAM arrays. Two different write-verify algorithms were considered and compared, namely, the incremental gate voltage with verify algorithm (IGVVA) and the incremental step pulse with verify algorithm (ISPVA). By using the IGVVA, a main tradeoff has to be considered since reducing the voltage step leads to a smaller DTD variability at the cost of a strong increase in energy consumption. Although the ISPVA cannot reduce the DTD variability as much as the IGVVA, its voltage step can be decreased in order to reduce the energy consumption with almost no impact on the DTD variability. Therefore, the final decision on which algorithm to employ should be based on the specific application targeted for the RRAM array.

Index Terms—Energy consumption, multilevel, programming algorithm, resistive random access memory (RRAM), voltage step.

I. INTRODUCTION

RESISTIVE random access memory (RRAM) technology based on metal-oxide dielectrics is one of the most promising candidates for implementing the next generation of nonvolatile memories (NVMs) as well as the synaptic weights integrated into hardware-based artificial neural networks (ANNs) [1], [2]. Among other reasons, RRAM devices have shown great potential due to its capability of defining

multiple conductive levels in one single device [multilevel cell (MLC)], which allows the introduction of the multibits per cell paradigm on NVM applications and the quasi-analog modulation of synaptic weights on ANNs [3]–[8]. Since the creation and disruption of the conductive filament (CF), which is responsible for the switching behavior, is inherently stochastic [9], it is mandatory to use optimized programming algorithms to control both device-to-device (DTD) and cycle-to-cycle (CTC) variabilities [12]. The former refers to spatial variations of the switching behavior over the RRAM array mainly due to inhomogeneities of the fabrication process, whereas the latter refers to temporal variations on one device along switching cycles mainly due to the stochastic dynamics of the oxygen vacancies constituents of the CF [13]. One of the most successful approaches is the family of the so-called write-verify algorithms [10]–[12], [14]. In particular, the incremental step pulse with verify algorithm (ISPVA) has shown MLC capabilities with the definition of multiple low-resistive states (LRSs) on one-transistor/one-resistor (1T1R) structures by tuning two parameters during the set operation, namely, the target current (I_{trg}) and the gate voltage (V_G), which defines the compliance current [16]. The ISPVA applies a sequence of increasing voltage amplitude pulses on the top electrode (TE) terminal of the RRAM device. In recent years, an alternative emerged as a really promising approach, that is, instead of applying a constant V_G and a sequence of increased voltage amplitude pulses on the TE terminal during the set operation, the incremental gate voltage with verify algorithm (IGVVA) applies an incremental sequence of V_G values on the gate terminal, while it keeps constant the amplitude of the sequence of pulses applied on the TE terminal [17], [18].

In this work, the impact of the voltage step (10, 20, 50, and 100 mV) used on the V_G and V_{TE} sequences during IGVVA and ISPVA programming operations, respectively, is assessed in terms of the DTD variability achieved on the readout current distributions of four different LRSs (I_{trg} equal to 10, 20, 30, and 40 μA , respectively) and in terms of the energy consumption featured by using batches of 1k devices integrated on 4-kbit 1T1R HfO₂-based RRAM arrays.

II. EXPERIMENTAL SETUP AND PROGRAMMING ALGORITHMS

The devices employed to test experimentally the impact of the V_G and V_{TE} step value used, respectively, on IGVVA

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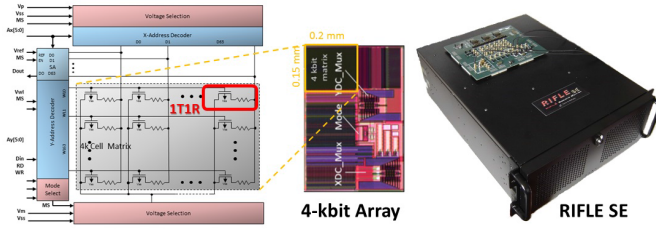


Fig. 1. Schematic and micrograph of the 4-kbit array (left) and picture of the electrical characterization setup: RIFLE SE (right).

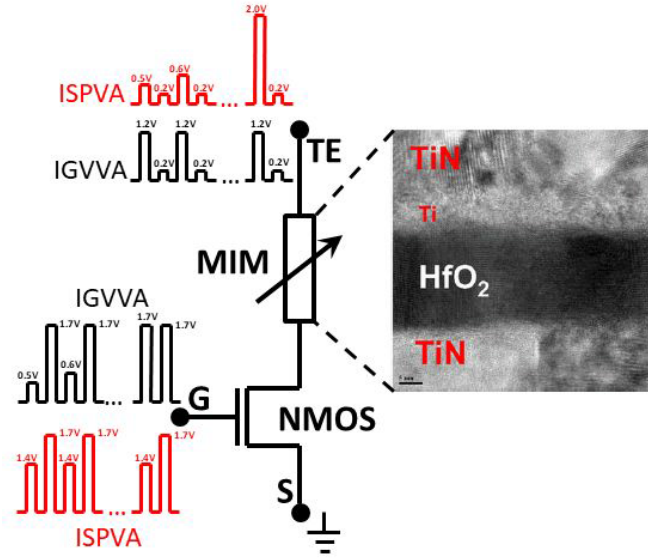


Fig. 2. Circuit schematic of the 1T1R cells (G: gate terminal; S: source terminal; and TE: top electrode terminal) with the voltage waveforms applied on each terminal during IGVVA (black) and ISPVA (red) programming operations (left) and cross-sectional TEM image of the MIM stack (right).

and ISPVA programming operations in order to define four discrete LRSs are RRAM cells integrated into 4-kbit arrays (see Fig. 1). These RRAM cells are constituted by an nMOS transistor (manufactured in 0.25- μm CMOS technology) connected in series to a metal–insulator–metal (MIM) structure (see Fig. 2) placed on metal line 2 of the CMOS process. The MIM structure consists of a TiN/HfO₂/Ti/TiN stack (see Fig. 2) with 150-nm TiN top and bottom electrodes deposited by magnetron sputtering, a 7-nm Ti layer (under the TiN TE), and a 8-nm HfO₂ switching layer grown by atomic layer deposition (ALD) [19]. After patterning the MIM structure with an area of about 0.4 μm^2 , a SiON layer was deposited to encapsulate the RRAM cell.

The strategy followed by the algorithms assessed in this work, namely, the IGVVA and the ISPVA, consists of applying sequences of voltage pulses on the gate and TE terminals of the RRAM device, while the source terminal is grounded (see Fig. 2) [18], [20]. Both algorithms were implemented and applied on the RRAM array by using the characterization setup shown in Fig. 1, namely, the RIFLE SE. For the IGVVA, the sequence of pulses applied on the gate terminal features a constant increment in amplitude from 0.5 to 1.7 V in steps

of 10, 20, 50, or 100 mV. The amplitude of the voltage pulses applied on the TE terminal is kept constant at 1.2 V during the programming stage. In contrast, for the ISPVA, the amplitude of the voltage pulses applied on the gate terminal is kept constant at a specific V_G value depending on the LRS programmed (1.0, 1.2, 1.4, or 1.6 V) [16]. In this case, the sequence of pulses applied on the TE terminal features a constant increment in amplitude from 0.5 to 2.0 V in steps of 10, 20, 50, or 100 mV. In both algorithms, after every programming pulse, a readout operation of the current flowing through the RRAM device is performed. The voltage values applied to perform this operation on the TE and gate terminals, respectively, are 0.2 and 1.7 V. Voltage pulses are applied on a specific RRAM device until either the readout current measured overcomes the I_{trg} value (10, 20, 30, or 40 μA) associated with the LRS targeted or the maximum amplitude in the voltage sequence is achieved.

III. IMPACT OF VOLTAGE STEP ON DTD VARIABILITY

In order to create a stable CF that activates the resistive switching (RS) behavior in every RRAM cell characterized in this study, the forming operation was performed in the three-step manner reported in [21]. First of all, a soft breakdown was performed by using the ISPVA with a voltage amplitude sweep on the TE terminal from 2.0 to 5.0 V in steps of 0.01 V and a pulsewidth equal to 10 μs . Afterward, the crucial first reset operation was carried out also by using the ISPVA with a voltage amplitude sweep on the source terminal from 0.5 to 2.0 V in steps of 0.1 V and with a pulsewidth equal to 1 μs .

Once the reliable RS behavior is established, each batch of 1k RRAM devices is programmed in one of the four desired LRSs, namely, LRS1, LRS2, LRS3, and LRS4 by tuning the I_{trg} parameter to 10, 20, 30, and 40 μA , respectively. For the IGVVA, the compliance current control is performed step-by-step by sweeping the V_G parameter from 0.5 to 1.7 V in steps of 10, 20, 50, or 100 mV. The pulses applied on the TE terminal feature a constant amplitude of 1.2 V and a pulsewidth of 1 μs . In contrast, for the ISPVA, the compliance current control is performed by defining a constant value for the V_G parameter related to the LRS targeted, namely, 1.0, 1.2, 1.4, and 1.6 V for LRS1, LRS2, LRS3, and LRS4, respectively. The pulses applied on the TE terminal feature a constant increase in amplitude from 0.5 to 2.0 V in steps of 10, 20, 50, or 100 mV and a pulsewidth of 1 μs . An in-depth study of the impact of the programming pulsewidth on the switching performance of RRAM devices can be found in [22].

The resulting cumulative distribution functions (CDFs) of the readout currents measured by the IGVVA and the ISPVA just after the set transition for the four different LRSs by using the four different voltage steps (V_{G_step} and V_{TE_step} , respectively) are shown in Figs. 3(a) and 4(a), respectively. According to the results obtained by using the IGVVA, there are two trends to highlight. First of all, there is a slight reduction on the dispersion of the current CDFs by increasing the I_{trg} value [23]. Second, by decreasing the V_{G_step} value, the current CDFs are significantly straightened toward the

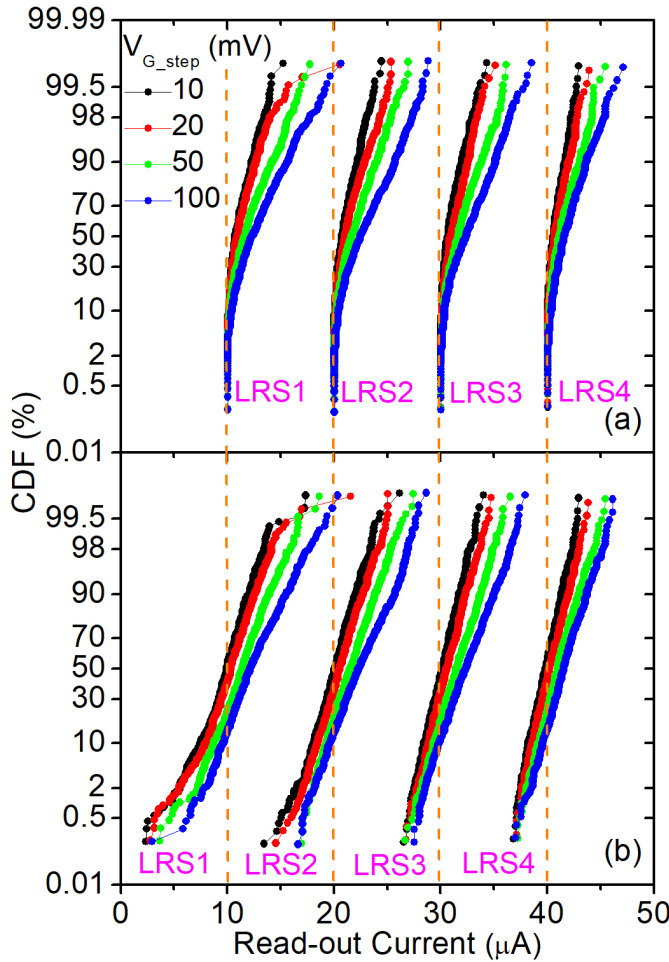


Fig. 3. CDFs of the readout currents measured by the IGVA (a) just after the set transition and (b) at the end of the programming operation for the four LRSs: LRS1, LRS2, LRS3, and LRS4 by using the four V_{G_step} values: 10, 20, 50, and 100 mV.

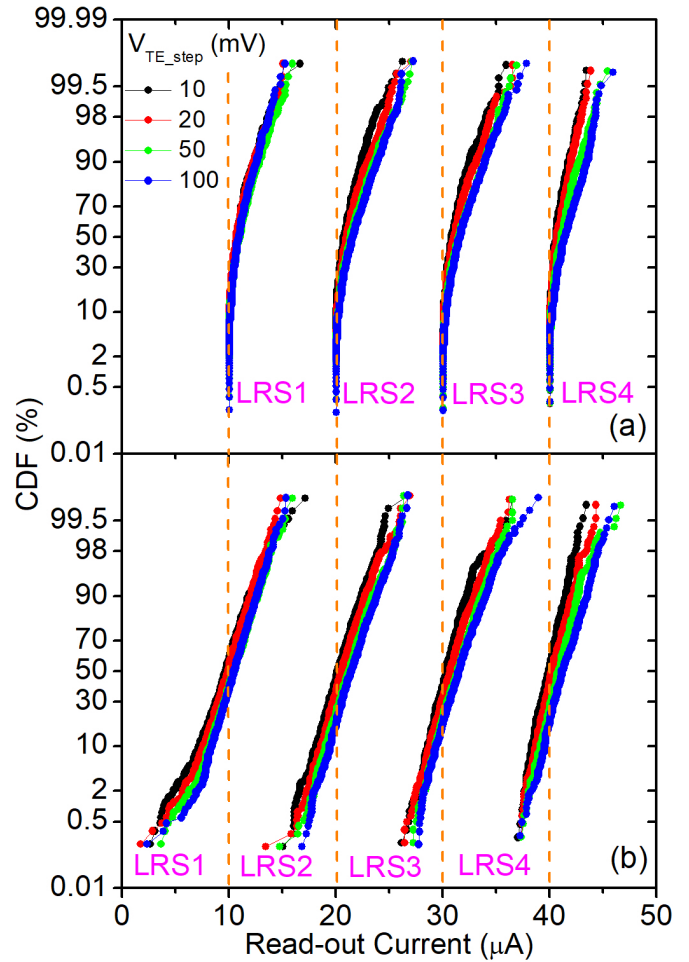


Fig. 4. CDFs of the readout currents measured by the ISPVA (a) just after the set transition and (b) at the end of the programming operation for the four LRSs: LRS1, LRS2, LRS3, and LRS4 by using the four V_{TE_step} values: 10, 20, 50, and 100 mV.

I_{trg} values, which enlarges the gap between LRS levels, thus improving the throughput of the IGVA. However, by using the ISPVA, these two trends practically vanish. There is no clear reduction on the dispersion of the CDFs by increasing I_{trg} . Moreover, by decreasing the V_{TE_step} value, the current CDFs show just a slight reduction on the dispersion. As a result, the ISPVA shows the best results for programming LRS1 regardless of the voltage step used. For the other three LRSs, by using a voltage step of 10 mV, the lowest variability is achieved by the IGVA, whereas by using a voltage step of 100 mV, the lowest variability is achieved by the ISPVA. By using voltage steps of 20 and 50 mV, there is no significant difference between the ISPVA and the IGVA.

An architectural constraint of the test setup imposes that the read-verify operations have to be continuously performed on the whole batch of RRAM cells until the algorithm finishes. The readout current measured during the last read-verify operation is also saved by the test setup. This leads to the possibility of studying the postprogramming perturbations of the LRSs defined on the RRAM devices under test [24]. Such perturbations are appreciable in Figs. 3(b) and 4(b) for the IGVA and the ISPVA, respectively, as tails of the current

CDFs below the corresponding I_{trg} value. The postprogramming relaxation of the CDFs is less pronounced for more conductive LRSs (in line with the data retention behavior reported in [16]), whereas a lower value of the voltage step leads to a bigger percentage of cells that cross down the I_{trg} limit of its LRS. Therefore, there is a tradeoff when choosing the optimal voltage step value (V_{G_step} and V_{TE_step}) in order to implement MLC on RRAM devices by using the IGVA and the ISPVA approaches. A smaller voltage step value leads in principle to a smaller DTD variability of programmed values on RRAM devices at the cost of a stronger tailing of the current CDFs below the I_{trg} limit at the end of the programming operation. However, in particular for the IGVA, according to Fig. 3(b), the gap between LRS levels remains slightly wider (especially between LRS2-LRS3 and LRS3-LRS4) by using smaller V_{G_step} values.

Figs. 5 and 6 show how the tailing of the current CDFs takes place by correlating DTD the readout current values measured just after the set transition and at the end of the algorithm. Most of the RRAM devices suffer from slight perturbations in their stored value, while those RRAM devices with stored values close to the I_{trg} limit are more prone to large variations.

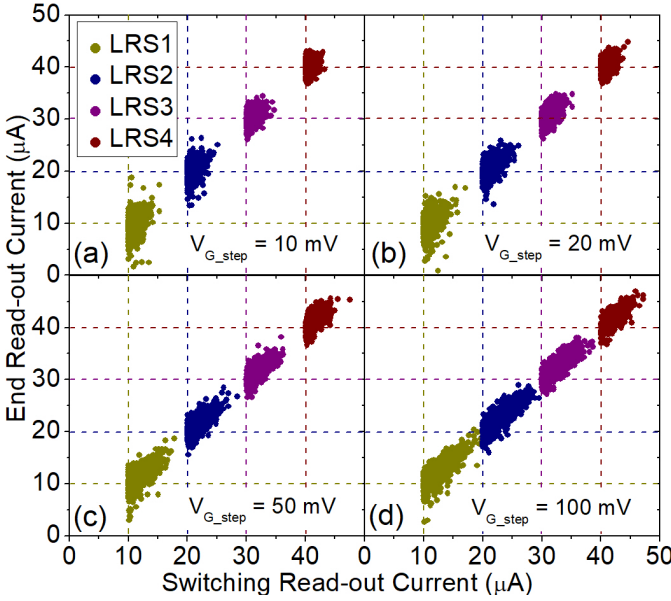


Fig. 5. DTD correlation of the readout currents measured just after the set transition and at the end of the IGVVA for the four LRS levels by using (a) 10, (b) 20, (c) 50, and (d) 100 mV as V_{G_step} values.

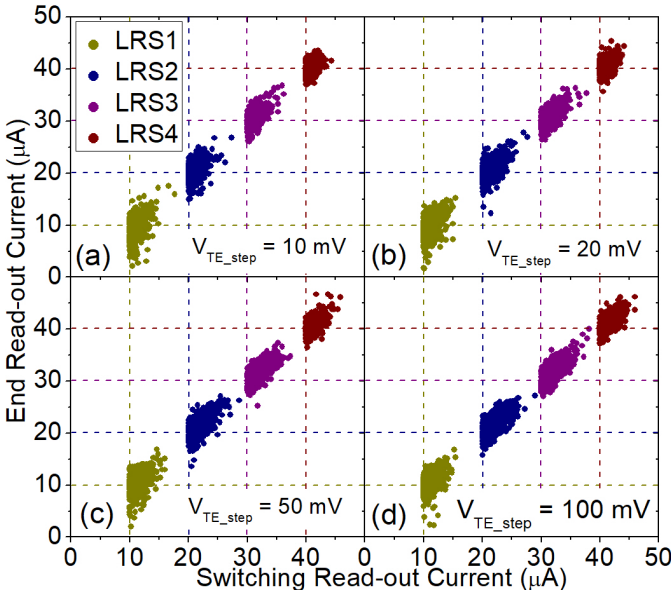


Fig. 6. DTD correlation of the readout currents measured just after the set transition and at the end of the ISPVA for the four LRS levels by using (a) 10, (b) 20, (c) 50, and (d) 100 mV as V_{TE_step} values.

In particular, narrower CDFs achieved by using smaller voltage step values are more populated close to I_{trg} , and therefore, a larger amount of RRAM devices cross down this limit due to the postprogramming perturbations.

Figs. 7 and 8 show the dependence of V_G and V_{TE} values at which the set switching takes place on the voltage step value used during the IGVVA and the ISPVA programming, respectively, for the four different LRSs. In case of programming with the IGVVA, the switching V_G values increase when targeting a larger I_{trg} since the maximum current flowing through the RRAM device is directly controlled by the gate voltage

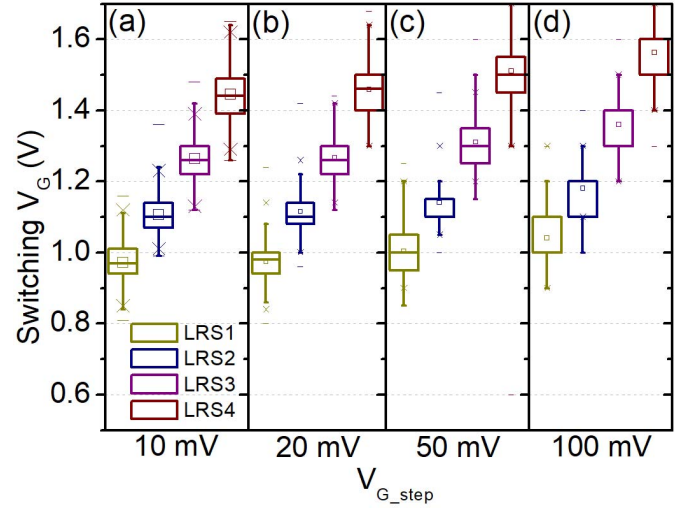


Fig. 7. Box plots of switching V_G values for the four different LRSs by using the IGVVA with V_{G_step} equal to (a) 10, (b) 20, (c) 50, and (d) 100 mV.

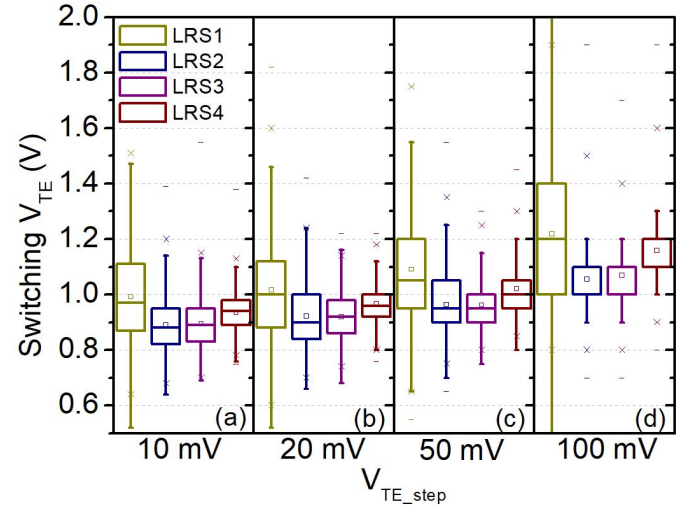


Fig. 8. Box plots of switching V_{TE} values for the four different LRSs by using the ISPVA with V_{TE_step} equal to (a) 10, (b) 20, (c) 50, and (d) 100 mV.

applied on the transistor. Hence, the number of programming pulses applied by the IGVVA until the set transition takes place increases linearly with the I_{trg} value associated with the targeted LRS, as shown in Fig. 9(a). The switching V_G distributions are quite similar regardless of V_{G_step} employed. Only a small growth can be noticed with rising V_{G_step} . This growth is due to the fact that the use of larger V_{G_step} values skips intermediate V_G values [see Fig. 9(a)] that could be the switching ones; however, they will never be applied. Therefore, the switching event takes place at a V_G value larger than the one actually required by the RRAM cell. In addition, such switching transitions at larger V_G values than the necessary lead to RRAM devices programmed at larger conductivities and, therefore, to broader readout current CDFs by using a larger V_{G_step} , as already shown in Fig. 3(a). In case of programming with the ISPVA, there is no such a linear dependence between the V_{TE} values at which the set switching

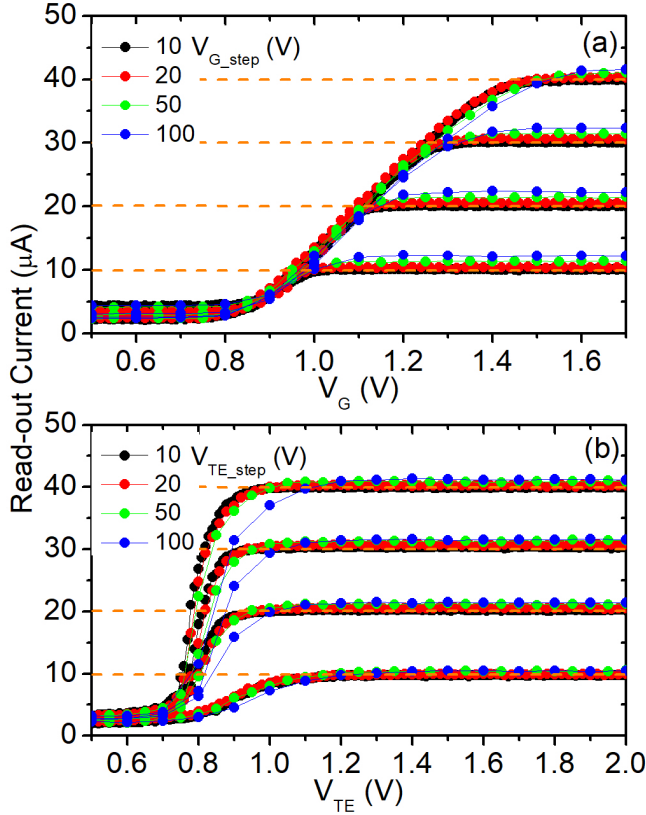


Fig. 9. Median values of the evolution of the readout currents measured during (a) IGVVA and (b) ISPVA programming by using 10, 20, 50, and 100 mV as V_{G_step} and V_{TE_step} values, respectively.

takes place and the LRS programmed. The transition from the high-resistive state (HRS) to an LRS with a higher I_{trg} value requires a larger current increase, whereas the larger V_G value applied on the transistor lets the current rise to be faster, as clearly shown in Fig. 9(b). The switching V_{TE} distributions also feature a small increase with rising V_{TE_step} . In this case, the increase may be due to the fact that by using a larger V_{TE_step} value, the number of voltage pulses applied to the RRAM sample is lower. Hence, the whole electrical stress needed to recreate the CF has to be compensated by applying voltage pulses on the TE terminal up to larger amplitudes [see Fig. 9(b)] [25]. It is worth noting that LRS1 shows a remarkable dispersion in all its corresponding switching V_{TE} distributions. A V_G value of 1.0 V may be a bit too low for a I_{trg} value equal to 10 μA , as shown in Fig. 7 [16].

IV. IMPACT OF VOLTAGE STEP ON ENERGY CONSUMPTION

Finally, the energy consumption featured during both IGVVA and ISPVA set operations was assessed for the four LRSs by using the four voltage step values. The average amount of energy required to carry out a set operation on a single RRAM device has been estimated according to the details explained in [22] and following the equation:

$$E = \sum_{i=1}^N V_{pr_i} \cdot I_{pr_i} \cdot T_{pr} + V_{rd} \cdot I_{rd_i} \cdot T_{rd} \quad (1)$$

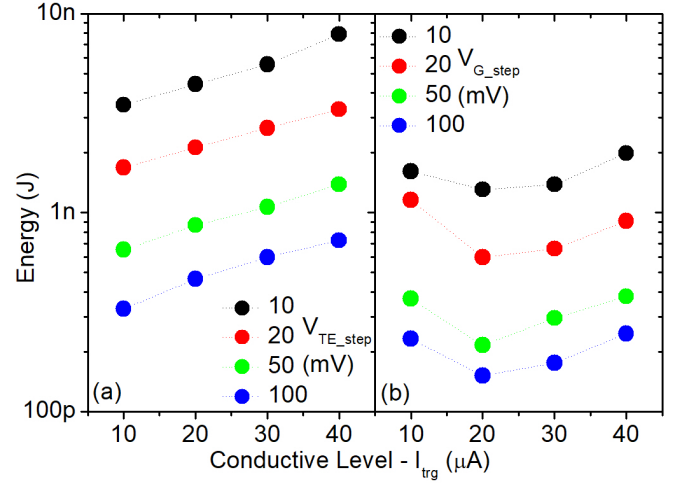


Fig. 10. Average energy required to perform multilevel set operations on a single RRAM cell for the four LRSs: LRS1 (10 μA), LRS2 (20 μA), LRS3 (30 μA), and LRS4 (40 μA)—by using the four V_{G_step} and V_{TE_step} values: 10, 20, 50, and 100 mV—with (a) IGVVA and (b) ISPVA.

where N is the average number of steps carried out during the set operation, V_{pr_i} is the voltage amplitude of the programming pulse applied at step i (equal to a constant value of 1.2 V for the IGVVA and increasing from 0.5 to 2.0 V for the ISPVA), I_{pr_i} is the current flowing through the RRAM cell at the programming step i , V_{rd} is the voltage amplitude of the readout pulse equal to 0.2 V, I_{rd_i} is the current measured through the readout operation at step i , and T_{pr} and T_{rd} are the pulsewidths of the programming and the readout pulses, respectively, both equal to 1 μs .

Fig. 10(a) shows a growth of the energy consumption with rising LRS conductance, in line with the increase of the number of pulses required by the IGVVA shown in Fig. 9(a). This dependence is strongly reduced for the ISPVA in Fig. 10(b) and mainly governed by the distribution of switching V_{TE} values shown in Fig. 8. In addition, the energy consumption featured by the ISPVA is remarkably smaller than that by the IGVVA in equivalent conditions (the same LRS and voltage step value). Much more significant is the energy consumption growth associated with the reduction of the voltage step value. For both algorithms, an increase of about one order of magnitude takes place between 100- and 10-mV voltage steps. Such an increase is due to the fact that the reduction of V_{G_step} and V_{TE_step} values increments significantly the number of programming and readout pulses needed to be applied on the RRAM cells during a set operation, as shown in Fig. 9.

These results lead to an additional tradeoff where the DTD variability of the current CDFs together with the energy consumption must be considered in order to implement the optimal programming approach, namely, IGVVA or ISPVA, together with the optimal voltage step for the specific application in which the RRAM array will be involved. By working with the ISPVA, it seems like the DTD variability reduction achieved by increasing V_{TE_step} that is not sufficient to justify the growth of the energy consumption. In contrast, the use of $V_{G_step} = 10$ mV with the IGVVA features the largest energy

consumption with the advantage of achieving the lowest DTD variability.

V. CONCLUSION

In this study, the impact of the voltage step value (V_{G_step} and V_{TE_step}) implemented on the programming algorithm (IGVVA and ISPVA, respectively) was assessed in an MLC context with four LRS levels. Despite the tailing of the current CDFs due to postprogramming instabilities, employing a smaller V_{G_step} value with the IGVVA still provides a better performance in terms of DTD variability. However, this reduction of the V_{G_step} value leads to a significant increase in the energy consumed during each set programming operation. In case of using the ISPVA, the slight decrease of the DTD variability reached by reducing the V_{TE_step} value does not compensate for the increase featured by the energy consumption. Therefore, the energy consumption can be strongly decreased by using larger V_{TE_step} values without substantial deterioration of the DTD variability. These tradeoffs imply the need of taking a careful decision based on the final application targeted for the RRAM-based array to be programmed, e.g., in an RRAM-based ANN trained off-line, the IGVVA with a small V_{G_step} value may be the best option, whereas in an RRAM-based ANN trained on-line or in an RRAM-based NVM, the ISPVA with a larger V_{TE_step} value may work better.

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